OS & SP

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SKILL – 13

1.Skill13.c

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/wait.h>

int main(void)
{
   int input_fd;
   int output_fd;
   pid_t pid;

   /* Open input file */
```

```
input_fd = open("input.txt", O_RDONLY);

if (input_fd == -1) {
    perror("Error opening input.txt");
    return 1;
}

/* Create output file */
output_fd = open(
    "output.txt",
    O_WRONLY | O_CREAT | O_TRUNC,
    0644
);

if (output_fd == -1) {
    perror("Error opening output.txt");
    close(input_fd);
    return 1;
```

```
}

pid = fork();

if (pid == -1) {
    perror("fork");
    close(input_fd);
    close(output_fd);
    return 1;
}

/* Child process */
if (pid == 0) {
    /* Redirect stdin from input.txt */
    if (dup2(input_fd, STDIN_FILENO) == -1) {
        perror("dup2 stdin");
        _exit(1);
    }
```

```
    /* Redirect stdout to output.txt */
    if (dup2(output_fd, STDOUT_FILENO) == -1) {
        perror("dup2 stdout");
        _exit(1);
    }

    close(input_fd);
    close(output_fd);

    /* cat reads stdin and writes stdout */
    execlp("cat", "cat", (char *)NULL);

    perror("execlp");
    _exit(1);
}

/* Parent process */
```

```
    close(input_fd);
    close(output_fd);
    waitpid(pid, NULL, 0);
    printf("Input redirected from input.txt\n");
    printf("Output redirected to output.txt\n");
    return 0;
}
```

OUTPUT :

